Lecture # 19

Date: Oct 30, 25

Day: Thurs

Recurrence Relation of LCS

$$LCS(N, M) = \begin{cases} 0 & \text{IF } N == 0 \,\|\, M == 0 \\ 1 + LCS(N-1, M-1) & \text{IF } A[N] == B[M] \\ \max\begin{pmatrix} LCS(N-1, M), \\ LCS(N, M-1) \end{pmatrix} & \text{if } A[N] \;!= B[M] \end{cases}$$

Recurrence of 0/1 Knapsack:

$$KSP(N, M) = \begin{cases} 0 & \text{IF } N == 0 \,\|\, M == 0 \\ KSP(N-1, M) & \text{IF } M < W[N] \\ \max(KSP(N-1, M), \\ \quad KSP(N-1, M-W[N]) + V[N]) & \text{IF } M > W[N] \end{cases}$$

Recurrence of Coin Change:

$$CC(N, Amt) = \begin{cases} Amt & \text{if } N == 1 \\ 0 & \text{if } Amt == 0 \\ CC(N-1, Amt) & \text{if } C[N] > Amt \\ \min(CC(N-1, Amt), \\ \quad CC(N, Amt - C[N]) + 1 \end{cases}$$

Date: Day: 96

Count all possible ways to return the Amount

Amt = 28,000

$\boxed{500, 1000, 5000}$

$$CC = \begin{cases} 0 & \text{IF } N == 0 \\ 1 & \text{IF } Amt == 0 \\ CC(N-1, Amt) & \text{If } C[N] > Amt \\ CC(N-1, Amt) + CC(N, Amt - C[N]) & \text{else} \end{cases}$$